

AETHERA · SPECIALTY BILLING SERIES

Nephrology Billing Solutions

Maximizing Revenue Across Office E/M, Dialysis Management, Kidney Transplant, CKD Care, and Vascular Access

CPT 99202-99215 · 90945-90999 · 50300-50380 · 36818-36909 · G-codes (dialysis)

Prepared by: Aethera Healthcare Solutions · Lakeland, FL

Scope: CPT 99202-99215 · 90945-90999 · 50300-50380 · 36818-36909 · G-codes (dialysis)

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ABOUT AETHERA

Maximizing Revenue. Minimizing Burden.

Aethera Healthcare Solutions is your full-service medical billing partner headquartered in Lakeland, FL. We handle coding, claims, payments, denials, and collections — so you can focus on patients.

500+

Practices Served across 25+ specialties

\$12M+

Revenue Recovered for specialty clients

95%

Clean Claim Rate (vs 84% industry avg)

< 30 d

Average AR — fast cash conversion



Personal, Not Corporate

Dedicated account team with deep specialty and payer knowledge. You speak with the same experts every month — not a rotating call-center queue.



Performance Targets, In Writing

96%+ net collection · <5% denial rate · 95%+ coding accuracy · 24h payment posting · 48h claim submission · 65%+ appeal success rate.



Zero Risk To Start

No setup fees. No long-term contracts. 90-day cancel anytime. Free 5-day revenue cycle audit before you commit — keep the findings either way.

Nephrology edge: Aethera's dialysis G-code selection engine and vascular access PA tracker are purpose-built for nephrology — the two biggest revenue leak sources in the specialty.



NEPHROLOGY · MARKET CONTEXT

Why Nephrology Billing Matters Now

Nephrology is the highest-spend single-organ specialty in US healthcare — driven by the Medicare ESRD program (\$42B/yr), 808K Americans on dialysis, and ~130K waiting for kidney transplant. Monthly dialysis capitation, transplant bundling, and vascular access procedures create unique RCM complexity.

\$ 1 1 4 B 

US kidney disease treatment spend (2024)
Largest single-organ spend in US healthcare; ~37% of Medicare's disease-specific spend.

8 0 8 K 

Americans on dialysis (2024)
~130K additional patients waiting for kidney transplant at any time.

~ 1 8 % 

Nephrology claim denial rate (industry avg)
Driven by monthly dialysis G-code errors and vascular access PA denials.

\$ 4 2 B + 

Annual Medicare ESRD spend
Largest single Medicare disease category; covers dialysis, transplant, and immunosuppression.

Net effect: nephrology practices that master monthly dialysis G-code selection, vascular access PA workflow, and transplant + critical care documentation can recover **7 – 1 2 %** of total revenue currently lost to denials and under-billing.



NEPHROLOGY · OVERVIEW

The Four Pillars of Nephrology Billing

Each pillar represents a distinct service family with unique coding, modifier, and PA requirements.



01 · E/M + CK

Office E/M & Chronic Kidney Disease

Office visits (99202–99215), CCM (99490/99439), CKD staging with G-codes for dialysis patients. Recurring revenue from CCM monthly billing across CKD stages 1–5.

CPT 99202–99215 · 99490 · 99439
· G-codes



03 · TRANSF

Donor Nephrectomy, Recipient & Post-Transplant

Donor nephrectomy (50300–50320), recipient transplant (50360–50380), post-transplant critical care (99291–99292). Immunosuppression management + critical care time documentation required.

CPT 50300–50380 · 99291–99292



02 · DIALYSIS

Hemodialysis, Peritoneal & Monthly Capitation

Hemodialysis (90935–90947), peritoneal dialysis (90945), monthly CAPD/APD management (90951–90970). Monthly capitation billing — modality determines G-code selection.

CPT 90935–90999 · G-codes (dialysis) ·
Monthly capitation



04 · VASCULAR

AV Fistula, Graft, Catheter & Fistulagram

AV fistula creation (36818–36821), AV graft (36830), catheter placement (36901–36909), diagnostic fistulagram (36901). PA mandatory for elective access creation; modifier -26/-TC split for fistulagram interpretation.

CPT 36818–36909 · PA mandatory
· -26/-TC



PILLAR 01 · E/M + CKD CARE

Office Visits, CCM & CKD Staging

Nephrology office encounters cover CKD staging (stages 1–5), hypertension management, electrolyte disorders, pre-dialysis education, and post-transplant follow-up. The 2021 E/M coding reform shifted documentation to MDM-based. Chronic Care Management (CCM) is critical for nephrology patients — most have ≥2 chronic conditions (CKD + HTN + DM), qualifying for monthly CCM billing. G-codes apply once patient is on dialysis.

E/M coding (2021+ reform)

Code by Medical Decision Making complexity or time; 99202–99215. Modifier 25 required if same-day procedure (e.g. fistulagram).

Chronic Care Management (CCM)

99490 first 20 min/month (\$42); 99439 add'l 20 min (\$47); 99487 complex 60 min (\$100). Most CKD patients qualify (≥2 chronic conditions).

CKD staging & G-codes

Pre-dialysis CKD stages 1–5 use ICD-10 N18.1–N18.5. Once on dialysis, G-codes apply for monthly dialysis management (see Pillar 2).

CCM is the single most under-billed code in nephrology — most CKD stage 3–5 patients qualify but few practices enroll them. $\sim \$42/\text{month} \times 12 = \sim \$500/\text{year}$ per patient of recurring revenue at near-zero marginal cost.

PILLAR 01 · TOP CI

Office E/M, CCM & Critical Care

Code	Service	Avg \$
99213	Est. E/M, low MDM	\$92
99214	Est. E/M, moderate MDM	\$131
99215	Est. E/M, high MDM	\$187
99490	CCM, first 20 min/month	\$42
99439	CCM, add'l 20 min/month	\$47
99291	Critical care, first 20 min	\$288

CCM enrollment: 1,000-patient CKD panel generates $\sim \$500\text{K}/\text{year}$ of recurring monthly revenue at near-zero marginal cost.



PILLAR 02 · DIALYSIS MGMT

Hemodialysis, Peritoneal & Monthly Capitation

Dialysis management is the highest-complexity area of nephrology RCM. Monthly capitation codes (90951–90970) bundle all physician services for a dialysis patient for the month — modality (HD vs PD) and patient age determine G-code selection. Wrong code = full monthly denial.

1 Hemodialysis (inpatient)

CPT 90935 single physician eval (\$195); 90937 repeated eval (\$245). Used for hospitalized HD patients. Billed per evaluation, not monthly.

2 Hemodialysis & PD (other than monthly capitation)

CPT 90945 PD or other than HD, single (\$185); 90947 dialysis other than HD, repeated (\$245). Used for outpatient PD/Acute HD monthly management.

3 Monthly CAPD/APD Management

CPT 90951–90970 monthly capitation. 90951 CAPD < 20 yrs (\$325); 90970 APD ≥20 yrs (\$325). Age + modality (CAPD vs APD vs HD) drives code selection.

4 Monthly HD Management

CPT 90952–90962 HD monthly, age-tiered (\$300–\$325). Patient age: <2 yrs, 2–11, 12–19, ≥20 yrs. Four age bands × HD vs PD = 8 monthly capitation codes total.

Monthly Capitation Logic

One G-code per dialysis patient per month. Code is determined by (1) **modality**: HD, CAPD, APD; (2) **patient age band**: <2, 2–11, 12–19, ≥20. Wrong code = full monthly denial — most common nephrology billing error.

Frequency Rules

Daily HD (inpatient) = bill per eval (90935/90937). Outpatient monthly = ONE capitation code per month. Frequency errors trigger audit flags — daily HD billing requires medical necessity documentation.

Top PA / Coding Errors

(1) Wrong monthly G-code for dialysis modality, (2) Daily HD billing without medical necessity, (3) Coordination of benefits errors (Medicare secondary payer rules), (4) Missing physician attestation for monthly visit.



PILLAR 03 · TRANSPLANT

Donor Nephrectomy, Recipient Transplant & Post-Transplant Critical Care

Kidney transplant is the highest-revenue-per-case nephrology service line. Donor nephrectomy (living donor) and recipient transplant are surgical CPT codes (50300–50380) typically billed by transplant surgeon. Nephrologists manage pre-transplant evaluation, post-transplant immunosuppression, and critical care during the perioperative period. Critical care time documentation (99291–99292) is the highest-denial post-transplant code.

♥ Donor nephrectomy

50300 donor nephrectomy (living related, \$2,200); **50320** donor nephrectomy (cadaver, \$2,800). Billed by transplant surgeon; nephrologist manages pre-op donor workup.

+ Recipient transplant

50360 kidney transplant recipient (living donor, \$2,800); **50380** kidney transplant recipient (deceased donor, \$4,200). Nephrologist manages post-op immunosuppression + critical care.

♥ Post-transplant critical care

99291 critical care, first 30 min (\$288); **99292** each add'l 30 min (\$140). Perioperative transplant recipient often requires 1–3 days of critical care.

○ Immunosuppression management

Post-transplant immunosuppression managed via E/M (99213–99215) + drug J-codes (tacrolimus, mycophenolate, prednisone). Lifetime immunosuppression required.



Kidney transplant reimbursement range
Recipient — living vs deceased donor

\$2,200–\$4,200



Americans on kidney transplant waiting list
At any given time

~ 130 K



Critical care per 30-min increment
99291 first / 99292 add'l

\$288 + \$140



Per-transplant critical care revenue opportunity
2-day ICU stay, fully documented

\$1,500–\$3K

Critical care time documentation is the #1 post-transplant denial — 99291/99292 require minute-level time log. A 2-day transplant ICU stay can generate \$1,500–\$3,000 of critical care revenue if documented correctly — or \$0 if not.

PILLAR 04 · VASCULAR ACCESS

AV Fistula, Graft, Catheter & Fistulagram

Vascular access procedures are the surgical/procedural revenue line for nephrology — every dialysis patient needs durable access. AV fistula creation (36818–36821) is the gold standard; AV graft (36830) and tunneled catheter (36901–36909) are alternatives. PA mandatory for elective access creation; modifier -26/-TC split critical for fistulagram interpretation.

1 AV Fistula Creation

CPT 36818 (\$1,250) upper arm cephalic vein; 36819 (\$1,300) brachial vein transposition; 36821 (\$1,180) radiocephalic. Gold standard dialysis access — matures over 6–12 weeks. PA mandatory for elective creation.

2 AV Graft

CPT 36830 (\$1,650) AV graft, prosthetic (forearm loop or upper arm straight). Used when native fistula not feasible (small vessels, prior failures). PA required.

3 Catheter Placement

CPT 36901 (\$425) diagnostic fistulagram; 36902–36909 sequential intervention codes (angioplasty, stent, thrombectomy). Tunneled catheter placement 36558 (\$850).

4 Fistulagram & Intervention

CPT 36901 diagnostic fistulagram (\$425); 36903 angioplasty (\$650); 36905 stent placement (\$1,200); 36906 thrombectomy (\$950); 36907 access revision. Modifier -26/-TC split between vascular surgeon and radiologist.



AV fistula creation reimbursement
CPT 36818 — gold standard access

\$1,250



AV graft reimbursement
CPT 36830 — prosthetic alternative

\$1,650



Fistulagram + intervention range
CPT 36901–36909 diagnostic to stent

\$425+



PA workflow + modifier discipline



NEPHROLOGY • BILLING CODES

Top CPT/HCPCS Codes & 2026 Reimbursement

Most-billed codes across the four nephrology pillars.

Code	Service	Modifier	Avg \$	Notes
99213	Est. E/M, low MDM	—	\$92	Most common nephrology E/M
99214	Est. E/M, moderate MDM	—	\$131	CKD/hypertension management
99215	Est. E/M, high MDM	—	\$187	Complex CKD + comorbidities
90935	HD, single physician eval	—	\$195	Inpatient hemodialysis
90937	HD, repeated physician eval	—	\$245	Inpatient HD, repeated
90945	PD, single physician eval	—	\$185	Peritoneal dialysis, single
90947	Dialysis other than HD, repeated	—	\$245	PD/Acute HD, repeated
90951	CAPD monthly, <20 yrs	—	\$325	Monthly capitation, pediatric
90970	APD monthly, ≥20 yrs	—	\$325	Monthly capitation, adult APD

Modifier -26 — professional component (interpretation). **Modifier -TC** — technical component (equipment). **Modifier 25** — significant, separately identifiable E/M same day as procedure.

Monthly capitation (90951–90970) — one G-code per dialysis patient per month; modality (HD/CAPD/APD) + patient age band drive code selection. **Critical care (99291/99292)** — minute-level time log required.

50360	Kidney transplant, living donor recipient	—	\$2,800	Transplant surgeon
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NEPHROLOGY • DENIAL PREVENTION

Top 8 Nephrology Denial Reasons & How Aethera Prevents Each

Nephrology denial rates average 16–20%; each denied claim costs \$35–80 to rework. Monthly dialysis G-code errors and vascular access PA denials drive 50%+ of revenue loss.

RED Monthly dialysis management billing errors

28 %

Wrong G-code or 90951–90970 code selected for dialysis modality (HD vs CAPD vs APD) or patient age band.

AETHERADialysis G-code selection engine: modality + age + month → correct code, with monthly attestation.

RED Vascular access procedure PA missing

22 %

AV fistula/graft creation (36818/36830) billed without PA on file; full denial + rebilling delay.

AETHERAPA tracker auto-submits at surgical scheduling; blocks claim release until PA confirmed.

AMBER Modifier -26/-TC errors on fistulagram

13 %

Fistulagram interpretation (36901) billed with wrong modifier — both -26 and -TC, or modifier missing entirely.

AETHERAModifier rules engine: -26 for interpretation-only, -TC for facility, automatic per provider setup.

AMBER Transplant PA denied

11 %

Transplant evaluation documentation incomplete — missing pre-transplant workup or insurance authorization.

AETHERATransplant PA documentation checklist before submission; payor-specific eval templates.

AMBER CCM time documentation insufficient

9 %

99490/99439 billed without 20-minute time log supporting monthly threshold.

AETHERACCM time-tracking platform with auto-billing at 20-min threshold + audit-ready log.

GREEN Dialysis frequency exceeded

7 %

Daily HD billing (90935/90937) without medical necessity documentation supporting inpatient frequency.

AETHERAFrequency tracker enforces medical necessity documentation per CMS guidelines.

GREEN Critical care time documentation missing

6 %

99291/99292 billed without minute-level time log supporting critical care threshold.

AETHERACritical care time log: ICU encounter auto-prompts minute capture.

GREEN ESRD coordination of benefits errors

5 %

Medicare secondary payer rules misapplied — billing Medicare primary when commercial primary required.

AETHERACOB verification engine: Medicare ESRD rules (30-month coordination period) automated.



CASE STUDY · INTEGRATED NEPHROLOGY RCM

4-Nephrologist Practice — Annual Revenue Impact

A \$4.8M-revenue practice models the combined upside of dialysis G-code accuracy, vascular access PA recovery, and transplant + critical care documentation.

4 nephrologists \$4.8M Annual revenue 600 dialysis patients (480 HD, 120 PD) 40 transplants/year 240 vascular access procedures

Dialysis Mgmt Code Accuracy

COLUMN 01 · MONTHLY CAPITATION

- 600 dialysis patients (480 HD, 120 PD)
- Pre-Aethera error rate 18%
- Post-Aethera error rate 4%
- Recovered monthly claims 84/mo
- Avg monthly capitation value \$325

ANNUAL REVENUE

\$325,000

Vascular Access PA Recovery

COLUMN 02 · SURGICAL / PROCEDURAL

- 240 vascular access procedures/year
- Pre-Aethera PA denial rate 22%
- Post-Aethera PA denial rate 6%
- Recovered procedures 38/yr
- Avg procedure value (weighted) \$4,740

ANNUAL REVENUE

\$180,000

Transplant + Critical Care

COLUMN 03 · DOCUMENTATION

- 40 transplants/year + post-transplant CC
- Pre-Aethera CC denial rate 16%
- Post-Aethera CC denial rate 5%
- Recovered CC encounters 11/yr
- Avg CC value per transplant \$16,800

ANNUAL REVENUE

\$185,000

Combined annual revenue recovery: **\$690,000**. Implementation cost (year 1): ~ **\$182K** (Aethera RCM fees + dialysis G-code engine + PA workflow + critical care documentation platform). Year-1 net: **\$508,000**.

\$690K 280%
ANNUAL RECOVER RO

NEPHROLOGY · COMPLIANCE HOTSPOTS

Where OIG, Medicare RAC, and Commercial Payers Audit First

Top compliance risks across the four nephrology pillars. Quarterly self-audits recommended.

 E/M + CKD Care 0 1	 Dialysis Mgmt 0 2	 Transplant 0 3	 Vascular Access 0 4
RED CCM time documentation fraud — billing 99490/99439 without contemporaneous time logs totaling ≥20 minutes; OIG Work Plan 2025–2026 active audit target.	RED Wrong monthly G-code for dialysis modality — billing 90951 (CAPD) for HD patient or vice versa; full monthly denial + clawback risk.	RED Transplant without documented evaluation — billing 50360/50380 without complete pre-transplant workup documentation; payor clawback risk.	RED AV fistula/graft creation without PA — billing 36818/36830 without prior auth on file; full denial + surgical rebilling delay.
AMBER CKD stage documentation missing — billing E/M at high complexity (99215) without documented CKD stage (ICD-10 N18.1–N18.5) supporting MDM.	AMBER Dialysis frequency over-billed — billing daily HD (90935/90937) for outpatient patients on monthly capitation; double-billing violation.	AMBER Donor nephrectomy documentation insufficient — billing 50300/50320 without living donor consent + workup documentation.	AMBER Fistulagram modifier -26/-TC errors — billing both professional AND technical components or modifier missing entirely; auto-denial.
GREEN Modifier 25 misuse — appending to E/M when fistulagram on same day does not represent significant, separately identifiable service.	LOW Coordination of benefits errors — Medicare secondary payer rules misapplied during 30-month ESRD coordination period.	LOW Post-transplant critical care time over-billing — billing 99291/99292 beyond documented ICU minutes; time log audit required.	LOW Catheter placement documentation missing — billing 36901–36909 without fluoroscopy guidance documentation or operative note.

OIG Work Plan 2025–2026: explicitly lists dialysis G-code billing and CCM time logging as active audit targets. Medicare RACs have recovered > \$180M from nephrology practices since 2021 — dialysis monthly capitation errors are the #1 audit finding.



CLOSING • KEY TAKEAWAYS

Three Actions to Capture Nephrology Revenue

For nephrology practice administrators and revenue cycle directors leading the next 90 days.

01



Automate monthly dialysis G-code selection

Build EHR + billing system integration that auto-selects 90951–90970 monthly capitation code based on (1) dialysis modality (HD/CAPD/APD) and (2) patient age band (<2, 2–11, 12–19, ≥20). Add monthly physician attestation workflow. Dialysis G-code error rate drops from 18% to < 4% — recovering ~\$325K per 600-patient dialysis panel.

↗ \$325K/yr · 600-patient panel

02



Streamline vascular access PA workflow

Implement PA tracker that auto-submits at surgical scheduling for AV fistula (36818) and AV graft (36830) creation. Blocks claim release until PA confirmed. Add modifier -26/-TC rules engine for fistulagram (36901) split billing. Vascular access PA denial rate drops from 22% to < 6% — recovering ~\$180K per 240-procedure annual volume.

↗ \$180K/yr · 240 procedures

03



Enforce transplant + critical care time documentation

Implement critical care time log that auto-prompts minute capture during transplant ICU encounters (99291 first 30 min, 99292 each additional). Pre-submission rules engine blocks 99291/99292 claims without minute-level time log. Add transplant PA documentation checklist. Recovers ~\$185K per 40-transplant annual volume.

↗ \$185K/yr · 40 transplants

Questions & Discussion

Optimizing Nephrology Revenue Across Dialysis, Transplant, CKD & Vascular Access

✉ support@aetherahealthcare.co

@aetherahealthcare.com

☎ +1 (863) 694-0325 13